Safe Roads in Floods

A community-powered, open-source flood map with vehicle-aware safe routing

Tata Young Social Innovator Challenge | Sector: Disaster Resilience | Submitted by Vishnupriya V, B.E. ECE, RV Institute of Technology and Management, Bengaluru

The problem statement I am addressing

How can people know which roads are usable when a flood cuts off a neighbourhood?

1. The problem

When heavy rain floods a city, residents must decide within minutes whether a road is safe to cross, walk, ride or drive. The depth and extent of water on a particular street is rarely known to the person about to enter it. In Chennai, Cyclone Michaung in December 2023 left many neighbourhoods waterlogged and cut off, and people had to rely on word of mouth, social media posts and guesswork to find a way out or reach help.

The risks are practical and serious: vehicles stalling in deep water, people wading through unseen hazards, delayed access to hospitals, and delivery and relief workers losing time on blocked routes. The people most affected are those who cannot take risks: the elderly, people with disabilities, two-wheeler riders, and families without a vehicle.

2. The gap I found

General-purpose navigation apps are built to find the fastest route and are not designed around street-level flood depth for different vehicle types. Satellite radar can map large flooded areas through cloud, but it is not continuous: during the 2023 event only one Sentinel-1 satellite was operating (Sentinel-1B stopped in December 2021), so observations were infrequent, and radar is less reliable between buildings in dense cities. Neither source alone tells a resident which road to take right now, and that is the gap my project targets.

3. My solution in one paragraph

I propose an open-source web app where residents report flooded roads on a map with a simple depth category. Reports are confirmed by other users and expire automatically so old information never blocks a road. The app then computes a safe route for the user's vehicle type, and shows nearby shelters and hospitals. Satellite radar and elevation data add a flood-risk layer, and community reports fill the gaps between satellite passes.

4. Who it serves

- Residents of flood-prone neighbourhoods, including the elderly and people with disabilities
- Two-wheeler and car commuters, delivery and gig workers
- Volunteers, community groups and responders who need to know which roads are passable

5. How my system works

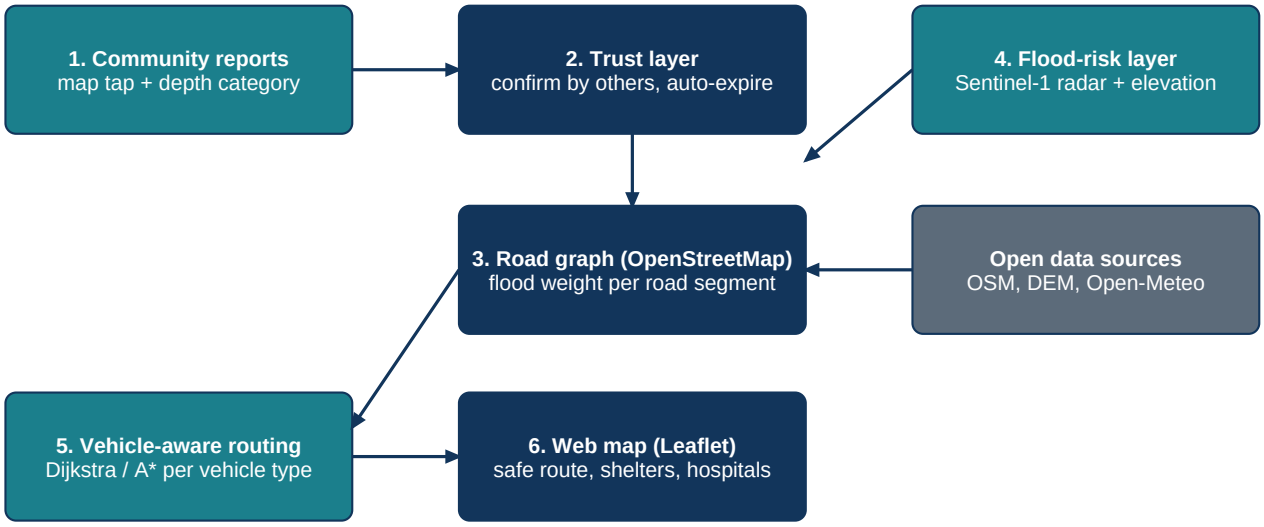

Figure 1. System architecture. Every component uses free, open-source tools or open data.

Depth categories and vehicle-aware routing

In my design, users choose from four reference levels instead of typing a number, so reports stay consistent. A flooded road segment is not simply closed; it receives a penalty that depends on depth and the user's vehicle. The thresholds below are my starting assumptions; I will make them adjustable and validate them against published guidance and local responders.

Reported depth	Two-wheeler	Car	High-clearance vehicle
Ankle	Avoid if alternative exists	Pass	Pass
Knee	Blocked	Avoid if alternative exists	Pass
Half wheel	Blocked	Blocked	Avoid if alternative exists
Above wheel	Blocked	Blocked	Blocked

Keeping the data trustworthy

- **Confirmation:** a report gains confidence when other users confirm it or report the same road.
- **Expiry:** reports fade after a set number of hours unless re-confirmed, because floodwater recedes.
- **Cross-check:** reports are compared with elevation (low-lying roads are more plausible) and, where available, radar flood extent.
- **Photos (future):** optional photo attachments for human verification; automatic depth estimation from photos is research-level and is kept as future work.

Technology stack (all free / open-source)

Need	Tool
Road network and shelters	OpenStreetMap, OSMnx
Routing	NetworkX (Dijkstra / A*) with custom flood weights
Flood-risk layer	Sentinel-1 SAR via Google Earth Engine (non-commercial), Copernicus DEM elevation
Rainfall context	Open-Meteo API
Front end and hosting	Leaflet.js, GitHub Pages, free-tier backend

6. Proof-of-concept plan: Chennai, December 2023

I will use the Chennai flood of December 2023 as a retrospective case study, not as a claim of a live system. Using open data, I will (a) build Chennai's road graph, (b) mark low-lying and previously reported flooded areas, (c) compare Sentinel-1 radar images from before and after the event, where a suitable pass is available, and (d) show how the same trip produces different safe routes for a two-wheeler and a car. Where radar coverage during the peak is missing, I will state that openly; that gap is exactly what community reporting is meant to fill.

7. Roadmap

Phase	Scope	Status
1. Core prototype	Report map, depth picker, confirmation and expiry, vehicle-aware routing for a pilot zone in Chennai	I will build this first, part-time, at about 1 to 1.5 hours a day
2. Risk layer	Sentinel-1 and elevation flood-prone layer; Chennai 2023 case study	Proof-of-concept notebook
3. Extensions	Photo verification, offline maps, SMS / WhatsApp reporting, partnerships with community groups and local authorities; possible integration with vehicle dashboards or insurance risk mapping	Future scope

8. Expected impact

- Fewer risky entries into floodwater, through clearer street-level information
- Faster safe access to shelters and hospitals, especially for people with limited mobility
- A free, open-source tool that any city, college or community group can adopt for its own area
- A reusable method that combines slow but wide-area satellite data with fast but sparse community data

9. Limitations and risks (stated honestly)

- **Adoption:** a crowdsourced system needs enough users; I would start in one neighbourhood with community groups.
- **Accuracy and misuse:** self-reported depth is approximate and reports can be wrong or malicious, which is why confirmation, expiry and cross-checks are central to the design.
- **Radar limits:** satellite observations are infrequent and less reliable in dense urban areas, so radar is used only as a supporting layer.
- **Connectivity:** floods often disrupt networks; offline caching and SMS reporting are planned extensions.
- **Safety:** the app gives guidance, not a guarantee, and will say so clearly. Vehicle thresholds need validation with local experts.
- **Scope:** this is a student-built prototype, not an official warning system.

10. About me

I am Vishnupriya V, a 2026 Electronics and Communication Engineering graduate from RV Institute of Technology and Management, Bengaluru, with project experience in VLSI physical design and analog IC design. I care about this problem and plan to build it as a long-term open-source project whether or not it is shortlisted, developing it in the open on GitHub (github.com/Vishnupriya280104). I would welcome mentorship, local data and community partners to help me test it properly.